23K-0012 (Raghib Rizwan Rabani)
23K-0023 (Muhammad Umar)
23K-0062 (Fahad Faheem Shah)
AI Engineering.

$t_1 = [1\ 0\ 1]$, $t_2 = [0\ 1\ 0]$, $t_3 = [1\ 1\ 0]$

$$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad W_K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad W_V = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

$$Q = X \cdot W_Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$K = X \cdot W_K = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$V = X \cdot W_V = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}$$

$$QK^T = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$$QK^T/\sqrt{d_k} = \begin{bmatrix} 0 & 1.414 & 1.414 \\ 1.414 & 0 & 0.707 \\ 1.414 & 0.707 & 1.414 \end{bmatrix}$$

CamScanner

$$\text{Softmax}\left(QK^T/\sqrt{dk}\right) = \begin{bmatrix} 0.108 & 0.445 & 0.445 \\ 0.576 & 0.140 & 0.254 \\ 0.401 & 0.197 & 0.401 \end{bmatrix}$$

$$\text{Softmax}\left(QK^T/\sqrt{dk}\right) \cdot V = \begin{bmatrix} 0.108 & 0.445 & 0.445 \\ 0.576 & 0.140 & 0.254 \\ 0.401 & 0.197 & 0.401 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.554 & 1.891 \\ 0.860 & 1.424 \\ 0.802 & 1.598 \end{bmatrix}$$

Now,

$$\text{mask} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

So, Applying mask:-

$$QK^T/\sqrt{dk} = \begin{bmatrix} 0 & -\infty & -\infty \\ 1.414 & 0 & -\infty \\ 1.414 & 0.707 & 1.414 \end{bmatrix}$$

$$\text{Softmax}\left(QK^T/\sqrt{dk}\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0.8044 & 0.1956 & 0 \\ 0.4011 & 0.1978 & 0.4011 \end{bmatrix}$$

$$\text{Softmax}\left(QK^T/\sqrt{d_k}\right).V = \begin{bmatrix} 1 & 1 \\ 0.804 & 1.195 \\ 0.802 & 1.598 \end{bmatrix}$$

Causal masking ensures that the model cannot access future tokens while predicting the next word. Without this restriction, the model might use information from upcoming words to make predictions. This allows tokens to focus on itself and the tokens before it. This makes the model learn the order of words and understand how to predict the next token based on the available context.